

# DSA0603 – Data Handling and Visualization

Slot B

## CO2 – Assessment Test 4 (AT1)

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### Question 1 — Healthcare Analytics

Interdisciplinary Integration: Healthcare + Medical Statistics + Data Visualization

#### Dataset

| Patient | Age | BMI  | Blood Pressure | Disease Severity |
|---------|-----|------|----------------|------------------|
| P1      | 24  | 19.8 | 110            | Low              |
| P2      | 32  | 24.5 | 118            | Medium           |
| P3      | 41  | 28.1 | 132            | High             |
| P4      | 48  | 30.3 | 145            | High             |
| P5      | 56  | 33.2 | 152            | Critical         |
| P6      | 65  | 36.5 | 168            | Critical         |
| P7      | 38  | 26.2 | 124            | Medium           |
| P8      | 29  | 22.1 | 116            | Low              |

#### Task

- Map Age to the x-axis.
- Map Blood Pressure to the y-axis.
- Represent Disease Severity using an appropriate color palette.
- Represent BMI using point size.
- Justify why the aesthetic mappings and scales effectively communicate health risk.

#### Answer — R Program (ggplot2)

```
library(ggplot2)

patients <- data.frame(
  Patient = c("P1", "P2", "P3", "P4", "P5", "P6", "P7", "P8"),
  Age      = c(24, 32, 41, 48, 56, 65, 38, 29),
  BMI      = c(19.8, 24.5, 28.1, 30.3, 33.2, 36.5, 26.2, 22.1),
  BP       = c(110, 118, 132, 145, 152, 168, 124, 116),
  Severity = factor(c("Low", "Medium", "High", "High",
                      "Critical", "Critical", "Medium", "Low"),
                    levels = c("Low", "Medium", "High", "Critical"))
)

ggplot(patients, aes(x = Age, y = BP, color = Severity, size = BMI)) +
  geom_point(alpha = 0.85, stroke = 0.6) +
  scale_color_manual(values = c("Low" = "#2ca02c", "Medium" = "#ffbf00",
                                "High" = "#ff7f0e", "Critical" = "#d62728")) +
  scale_size_continuous(range = c(3, 10)) +
  labs(title = "Patient Risk Profile: Age vs Blood Pressure",
       subtitle = "Color = Disease Severity, Size = BMI",
       x = "Age (years)", y = "Blood Pressure (mmHg)",
       color = "Disease Severity", size = "BMI") +
```

```
theme_minimal(base_size = 13)
```

## Resulting Visualization

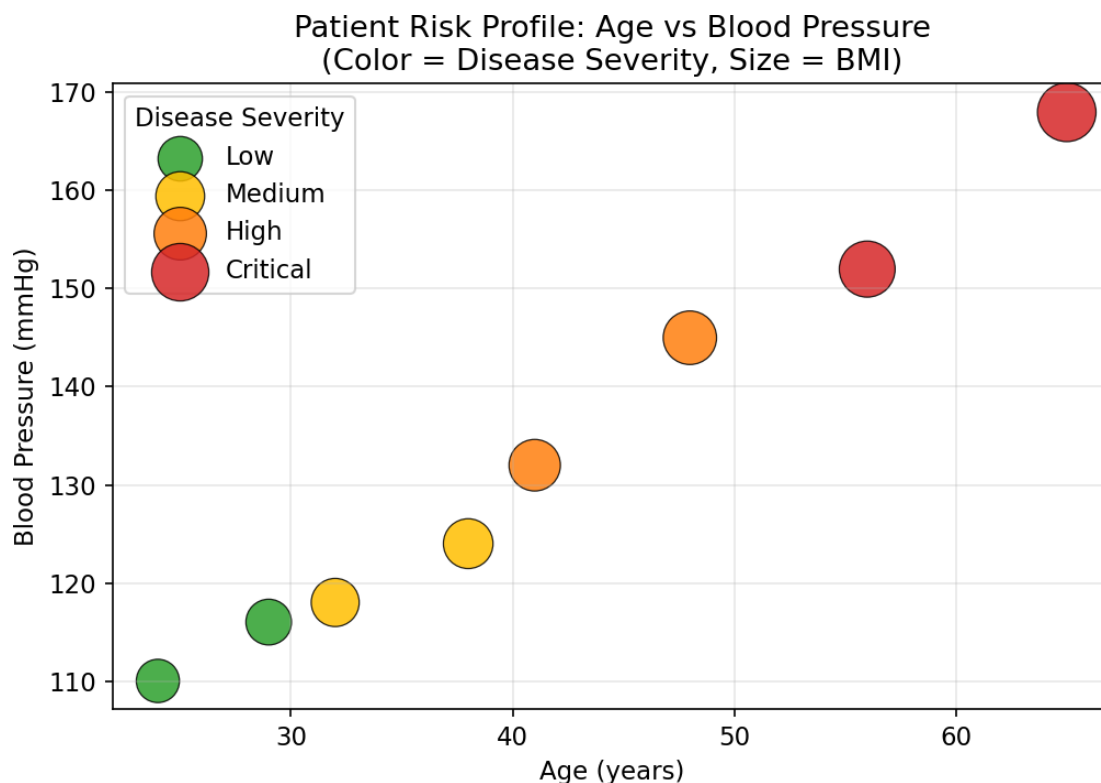


Fig 1. Age vs Blood Pressure scatter plot — color-coded by Disease Severity, sized by BMI.

## Justification

The x and y positions (Age, Blood Pressure) are the two variables most directly linked to cardiovascular risk, so placing them on the primary spatial axes lets the eye read the core clinical trend immediately: risk rises toward the upper-right. Color is used for Disease Severity because color is the strongest pre-attentive channel for categorical grouping — the ordered red-to-green palette (green = Low, amber = Medium, orange = High, red = Critical) mirrors the intuitive 'traffic-light' convention used in clinical dashboards, so a clinician can scan severity at a glance. Point size encodes BMI, a continuous secondary risk factor; size is well suited to continuous data and, combined with position and color, creates a three-variable-plus-category view without needing a fourth axis. Together the mappings let a viewer simultaneously assess how age, blood pressure, severity classification, and BMI co-vary — exactly the multi-factor pattern clinicians need to identify high-risk patients (older, higher BP, larger point, red color) versus low-risk patients (younger, lower BP, smaller point, green color).

## Question 2 — Climate Science

Interdisciplinary Integration: Environmental Science + Mathematics + Data Visualization

### Dataset

| Month | Solar Radiation | Temperature (°C) | Rainfall (mm) |
|-------|-----------------|------------------|---------------|
| Jan   | 180             | 20               | 25            |
| Feb   | 200             | 23               | 18            |
| Mar   | 240             | 29               | 12            |
| Apr   | 280             | 35               | 8             |
| May   | 320             | 39               | 5             |
| Jun   | 300             | 36               | 65            |
| Jul   | 260             | 32               | 150           |
| Aug   | 250             | 31               | 170           |
| Sep   | 240             | 30               | 120           |
| Oct   | 220             | 28               | 85            |
| Nov   | 190             | 24               | 45            |
| Dec   | 170             | 21               | 30            |

### Task

- Build a Cartesian line chart showing temperature variation across months.
- Build a Polar coordinate chart representing annual rainfall.
- Compare the advantages and limitations of Cartesian and Polar coordinate systems for climate visualization.

### Answer — R Program (ggplot2)

```
library(ggplot2)

climate <- data.frame(
  Month = factor(month.abb, levels = month.abb),
  Solar = c(180,200,240,280,320,300,260,250,240,220,190,170),
  Temp = c(20,23,29,35,39,36,32,31,30,28,24,21),
  Rain = c(25,18,12,8,5,65,150,170,120,85,45,30)
)

# 1. Cartesian line chart -- Temperature
ggplot(climate, aes(x = Month, y = Temp, group = 1)) +
  geom_line(color = "#d62728", linewidth = 1.1) +
  geom_point(color = "#d62728", size = 2) +
  labs(title = "Cartesian Line Chart: Monthly Temperature Variation",
       x = "Month", y = "Temperature (°C)") +
  theme_minimal(base_size = 13)

# 2. Polar coordinate chart -- Rainfall
ggplot(climate, aes(x = Month, y = Rain, fill = Rain)) +
  geom_col(width = 0.9, color = "black") +
  coord_polar() +
  scale_fill_gradient(low = "#deebf7", high = "#08519c") +
  labs(title = "Polar Chart: Annual Rainfall Distribution (mm)",
       x = NULL, y = NULL, fill = "Rainfall (mm)") +
  theme_minimal(base_size = 13)
```

### Resulting Visualizations

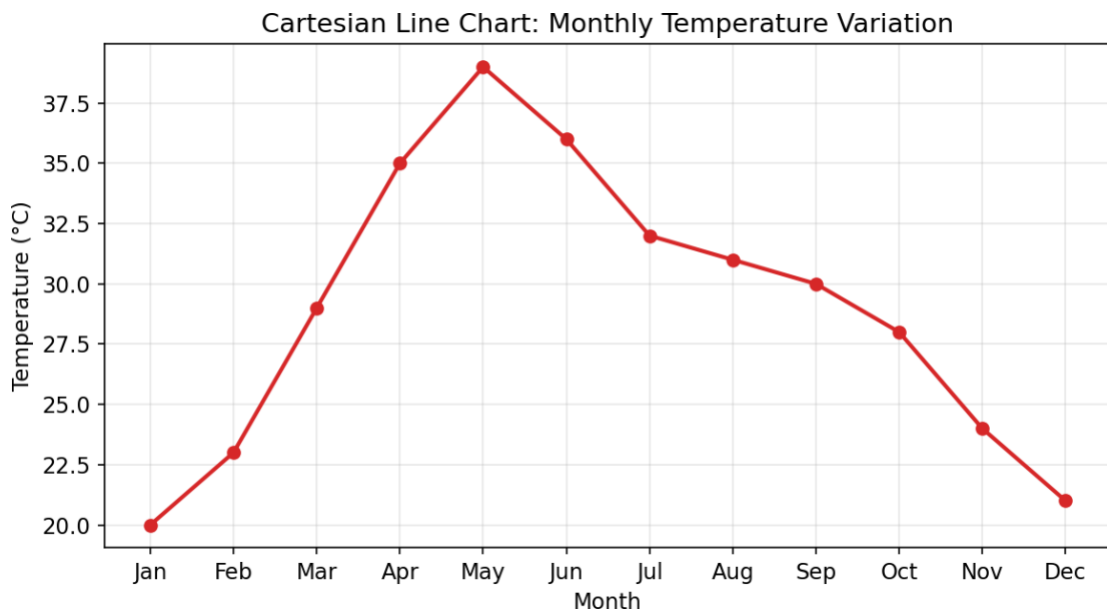


Fig 2a. Cartesian line chart — monthly temperature variation.

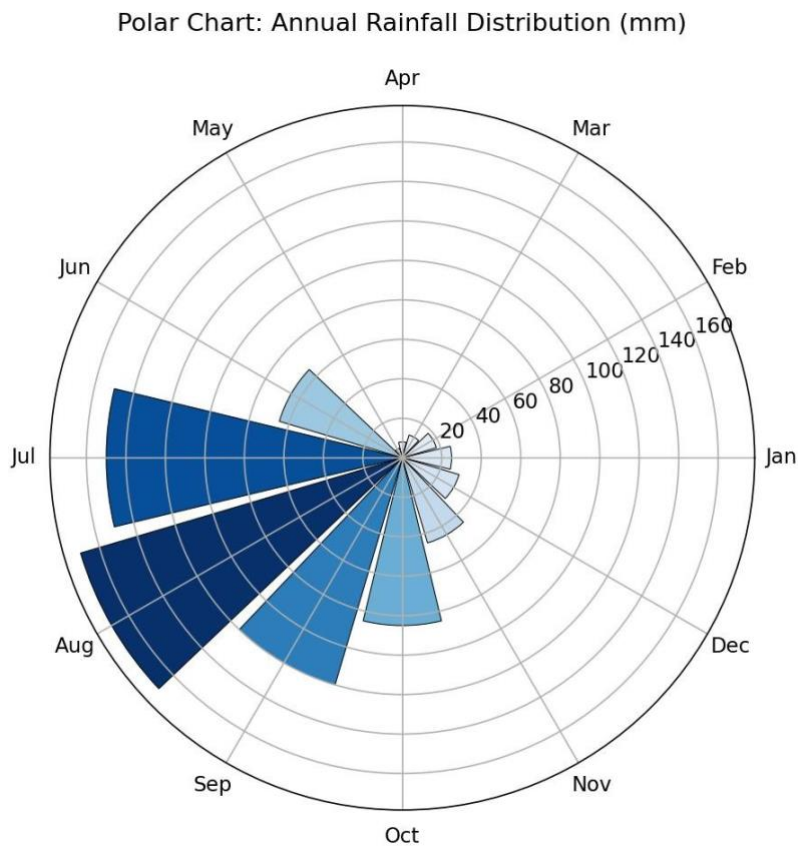


Fig 2b. Polar chart — annual rainfall distribution.

### Comparison — Cartesian vs Polar Coordinates

The Cartesian line chart is the better choice for temperature because temperature follows a smooth, continuous, non-cyclical-looking trend across the 12-month sequence, and a straight x-axis lets viewers read precise values, compare magnitudes accurately, and spot the rate of change (slope) between consecutive months without any angular distortion. Its main

limitation is that it does not visually convey the cyclical nature of the calendar — December and January appear far apart even though they are adjacent in the yearly cycle.

The Polar chart is well suited to rainfall because it naturally represents the cyclical, seasonal structure of a year — the monsoon peak (Jul–Aug) stands out as a visually distinct bulge, and the chart intuitively evokes a 'radar' or seasonal-wheel view that mirrors real-world seasonal recurrence. However, polar charts make precise value comparison harder: humans are less accurate at judging radial length and area than linear position, category labels get compressed near the center, and bars far from the origin can visually dominate even when the underlying difference in magnitude is modest. In practice, Cartesian charts are preferred for precise magnitude comparison and trend analysis, while Polar charts are preferred for emphasizing cyclical/seasonal patterns at the cost of reading accuracy.

## Question 3 — Business Analytics & Marketing

Interdisciplinary Integration: Marketing + Business Intelligence + Visualization

### Dataset

| Product | Sales (₹ Lakhs) | Profit (%) | Market Share (%) |
|---------|-----------------|------------|------------------|
| A       | 210             | 18         | 24               |
| B       | 185             | 14         | 20               |
| C       | 260             | 25         | 28               |
| D       | 145             | 12         | 15               |
| E       | 120             | 10         | 13               |

### Task

- Represent Sales using bar height.
- Represent Profit percentage using a sequential color scale.
- Highlight the product with the highest market share using a contrasting color.
- Explain why the chosen color palette improves business decision-making.

### Answer — R Program (ggplot2)

```
library(ggplot2)
library(dplyr)

sales_df <- data.frame(
  Product = c("A", "B", "C", "D", "E"),
  Sales   = c(210, 185, 260, 145, 120),
  Profit  = c(18, 14, 25, 12, 10),
  Share   = c(24, 20, 28, 15, 13)
)

top_share <- sales_df$Product[which.max(sales_df$Share)]
sales_df$Highlight <- ifelse(sales_df$Product == top_share, "Top Share", "Other")

ggplot(sales_df, aes(x = Product, y = Sales, fill = Profit)) +
  geom_col(color = "black") +
  geom_col(data = filter(sales_df, Highlight == "Top Share"),
    aes(x = Product, y = Sales), fill = NA, color = "#00b4d8", linewidth = 1.4) +
  scale_fill_gradient(low = "#ffffb2", high = "#bd0026") +
  labs(title = "Product Sales with Profit % Color Scale",
    subtitle = "Highest Market Share Product Highlighted in Blue",
    x = "Product", y = "Sales (₹ Lakhs)", fill = "Profit (%)") +
  theme_minimal(base_size = 13)
```

### Resulting Visualization

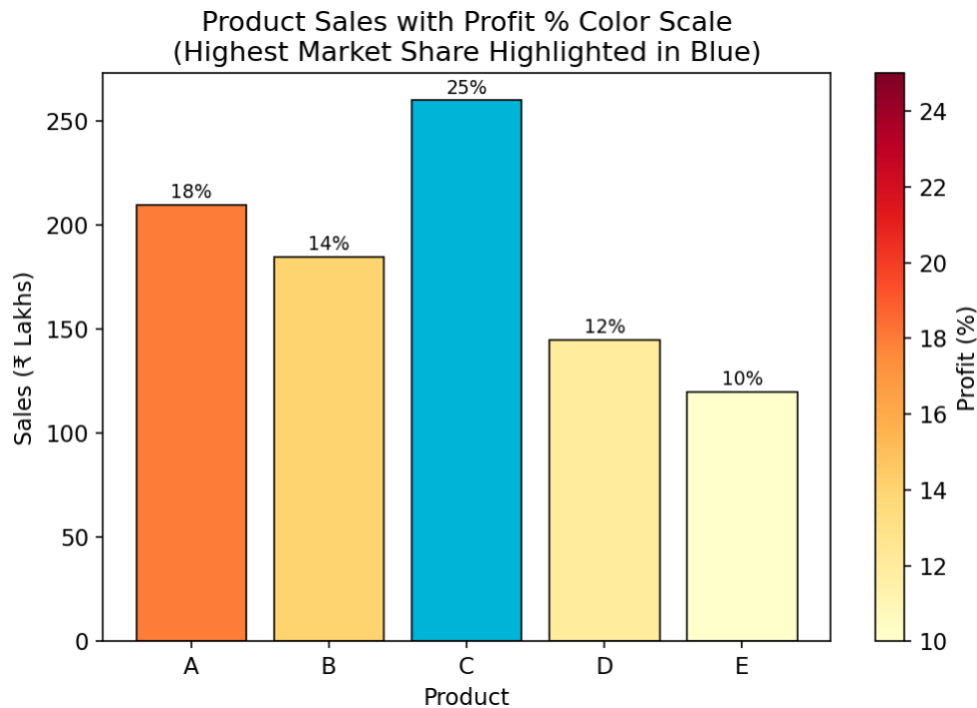


Fig 3. Sales by product — sequential color scale for Profit %, Product C (highest market share) highlighted.

#### Justification

Bar height is used for Sales because length/height is the most accurately perceived visual channel, letting decision-makers instantly rank products by revenue. A sequential (single-hue, light-to-dark) color scale is applied to Profit % because sequential palettes are designed for ordered, continuous data — the gradient shows at a glance not just which products earn more profit margin, but by how much, without implying arbitrary categorical distinctions. Highlighting Product C's market-share leadership with a contrasting outline color (rather than the sequential palette) ensures a single, unambiguous business signal — the star performer — stands out even though C is not the color-scale extreme on profit. This layered encoding lets a manager instantly identify (a) which products sell the most, (b) which are most profitable per unit sold, and (c) which one dominates the market, supporting fast, evidence-based decisions on where to invest marketing spend or renegotiate margins.

## Question 4 — Transportation Engineering

Interdisciplinary Integration: Civil Engineering + GIS + Transportation Analytics

### Dataset

| Direction  | Vehicles/hour |
|------------|---------------|
| North      | 980           |
| North-East | 760           |
| East       | 1240          |
| South-East | 890           |
| South      | 1420          |
| South-West | 1130          |
| West       | 860           |
| North-West | 720           |

### Task

- Construct a Polar Coordinate Chart representing traffic flow.
- Apply an appropriate color palette to distinguish traffic density.
- Explain why a Polar chart is more suitable than a Cartesian bar chart for directional traffic analysis.

### Answer — R Program (ggplot2)

```
library(ggplot2)

traffic <- data.frame(
  Direction = factor(c("North", "North-East", "East", "South-East",
                      "South", "South-West", "West", "North-West"),
                    levels = c("North", "North-East", "East", "South-East",
                              "South", "South-West", "West", "North-West")),
  Vehicles = c(980, 760, 1240, 890, 1420, 1130, 860, 720)
)

ggplot(traffic, aes(x = Direction, y = Vehicles, fill = Vehicles)) +
  geom_col(width = 0.85, color = "black") +
  coord_polar(start = 0) +
  scale_fill_viridis_c(option = "plasma") +
  labs(title = "Polar Chart: Directional Traffic Flow",
       x = NULL, y = "Vehicles/hour", fill = "Vehicles/hour") +
  theme_minimal(base_size = 13)
```

### Resulting Visualization

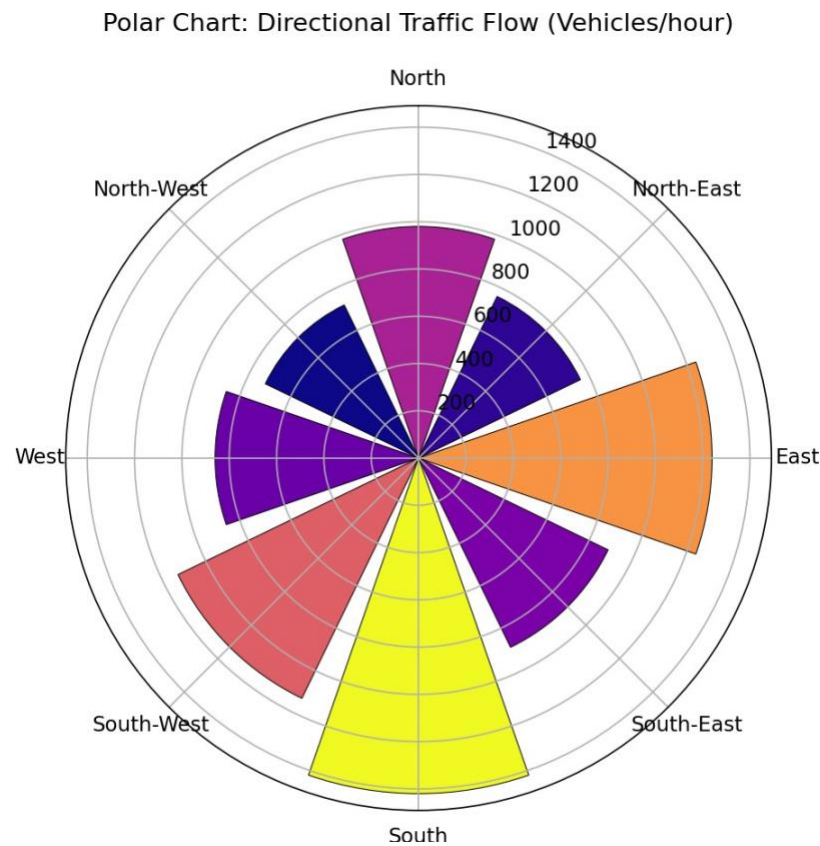


Fig 4. Polar chart of directional traffic flow at an eight-way intersection.

#### Justification

A Polar chart is more suitable than a Cartesian bar chart here because the underlying data is inherently directional — the eight categories (N, NE, E, SE, S, SW, W, NW) correspond to physical compass bearings around a real intersection, not an arbitrary ordered list. Plotting them radially preserves their true spatial relationship: North sits opposite South, East sits opposite West, exactly as in the physical road layout, so an engineer can instantly see which approach (e.g., South) carries the heaviest load and which opposing/adjacent directions are comparatively light. A Cartesian bar chart would force these directions into an arbitrary linear order, breaking the geometric intuition and making it harder to reason about opposite or adjacent approaches. The plasma color scale reinforces density perceptually — brighter, warmer bars mark the busiest approaches (South, East) — giving traffic planners an immediate, map-like read on congestion hot-spots for signal-timing or lane-widening decisions.

## Question 5 — Environmental Sustainability Dashboard

Interdisciplinary Integration: Environmental Science + Urban Planning + Data Visualization

### Dataset

| City | AQI | Green Cover (%) | CO <sub>2</sub> Emission | Population (Lakhs) |
|------|-----|-----------------|--------------------------|--------------------|
| A    | 48  | 45              | 1.8                      | 9                  |
| B    | 82  | 32              | 3.5                      | 15                 |
| C    | 135 | 21              | 5.6                      | 22                 |
| D    | 165 | 18              | 7.4                      | 28                 |
| E    | 72  | 38              | 2.9                      | 12                 |
| F    | 95  | 30              | 4.2                      | 18                 |

### Task

- Design a visualization dashboard with appropriate aesthetic mappings.
- Apply suitable scales for continuous variables.
- Use an effective color palette for AQI categories.
- Justify every visualization design choice.

### Answer — R Program (ggplot2 + patchwork)

```
library(ggplot2)
library(patchwork)

env <- data.frame(
  City   = c("A", "B", "C", "D", "E", "F"),
  AQI    = c(48, 82, 135, 165, 72, 95),
  Green  = c(45, 32, 21, 18, 38, 30),
  CO2    = c(1.8, 3.5, 5.6, 7.4, 2.9, 4.2),
  Pop    = c(9, 15, 22, 28, 12, 18)
)

env$AQI_Category <- cut(env$AQI, breaks = c(-Inf, 50, 100, 150, Inf),
  labels = c("Good", "Moderate", "Poor", "Severe"))
aqi_colors <- c(Good = "#2ca02c", Moderate = "#ffbf00",
  Poor = "#ff7f0e", Severe = "#d62728")

# Panel 1: Green cover vs CO2 emission (AQI category = color, Population = size)
p1 <- ggplot(env, aes(x = Green, y = CO2, color = AQI_Category, size = Pop)) +
  geom_point(alpha = 0.85) +
  geom_text(aes(label = City), vjust = -1.3, size = 3.5, show.legend = FALSE) +
  scale_color_manual(values = aqi_colors) +
  scale_size_continuous(range = c(4, 12)) +
  labs(title = "Green Cover vs CO2 Emission", x = "Green Cover (%)",
    y = "CO2 Emission (units)", color = "AQI Category", size = "Population (Lakhs)") +
  theme_minimal(base_size = 12)

# Panel 2: City-wise AQI bar chart with threshold reference lines
p2 <- ggplot(env, aes(x = City, y = AQI, fill = AQI_Category)) +
  geom_col(color = "black") +
  geom_hline(yintercept = c(50, 100, 150), linetype = "dashed", color = "grey40") +
  scale_fill_manual(values = aqi_colors) +
  labs(title = "City-wise Air Quality Index", x = "City", y = "AQI", fill = "AQI Category") +
  theme_minimal(base_size = 12)
```

## Resulting Dashboard

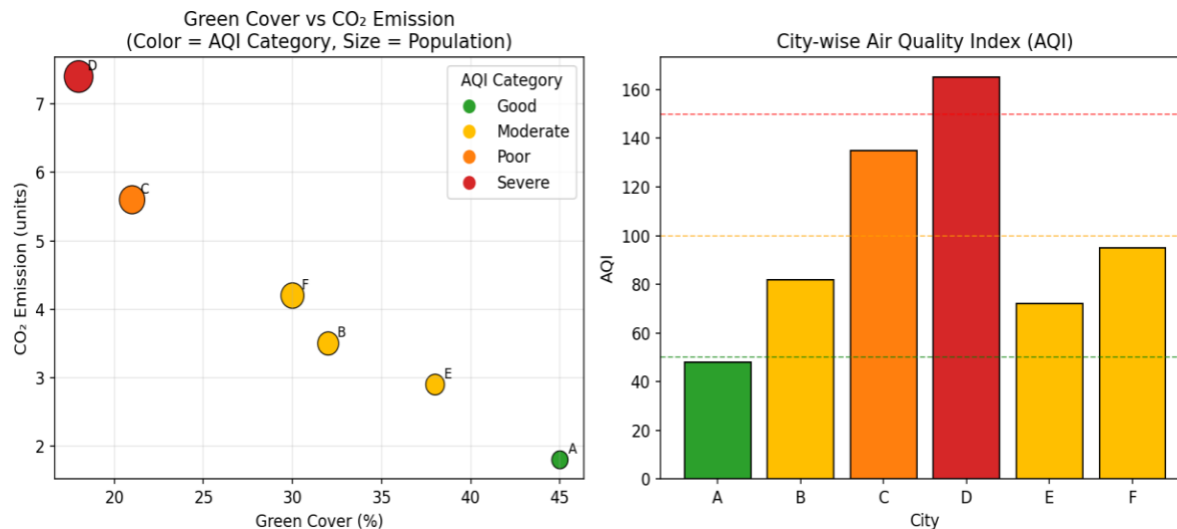


Fig 5. Two-panel sustainability dashboard — (left) Green Cover vs CO<sub>2</sub> Emission bubble chart, (right) City-wise AQI bar chart.

## Justification of Design Choices

- Aesthetic mappings: Green Cover and CO<sub>2</sub> Emission are placed on the x/y axes of the bubble chart because their trade-off (more green cover generally associates with lower emissions) is the core sustainability relationship the dashboard must communicate; position is the most precise perceptual channel, so this relationship is immediately visible.
- Color for AQI category: A discrete traffic-light-style palette (green → amber → orange → red for Good → Moderate → Poor → Severe) is used consistently across both panels because it matches a globally recognized air-quality convention, letting any viewer interpret severity without consulting a legend.
- Size for Population: Bubble size encodes population because larger populations amplify the human impact of poor air quality; combining size with color lets planners spot cities that are both heavily polluted and densely populated (e.g., City D) as the highest-priority intervention targets.
- Scales for continuous variables: Continuous linear scales are used for Green Cover, CO<sub>2</sub>, and AQI since all three are ratio-scale measurements where equal visual distances must represent equal real-world differences; a continuous size scale (not discrete bins) is used for population so that intermediate population values are still visually distinguishable.
- Dual-panel layout: The AQI bar chart complements the bubble chart by giving exact per-city AQI magnitudes (harder to read precisely from bubble color alone) with dashed threshold lines marking the Good/Moderate/Poor/Severe boundaries, so the dashboard supports both a fast relational read (bubble chart) and a precise magnitude read (bar chart).